Task 1: Introduction

ELK for Log Analysis and Investigations

In this room, we will learn how the Elastic Stack (ELK) can be used for log analysis and investigations. Although ELK is not a traditional SIEM, many SOC teams use it like one because of its data searching, and visualizing capability. We will explore how the components of ELK and learn how log analysis can be performed through it. We will also explore creating visualizations and dashboards in ELK.

The Elastic Stack (ELK)

① Elasticsearch → ② Logstash → ③ Kibana

① Elasticsearch
Distributed search and analytics engine. Stores and indexes all the data.

② Logstash
Data collection and processing pipeline. Ingests, parses and transforms logs.

③ Kibana
Data visualization and exploration interface. Create charts, dashboards and reports.

★ **Note:**
Although ELK is not a traditional SIEM, many SOC teams use it like one!

ELK helps SOC teams collect, search, analyze and visualize logs efficiently.

Learning Objectives

This room has the following learning objectives:

1. Understand the components of ELK and their use in SOC.
2. Explore the different features of ELK.
3. Learn to search and filter data in ELK.
4. Investigate VPN logs to identify anomalies.
5. Familiarize with creating visualizations and dashboards in ELK.

Key Takeaway ★

ELK is a powerful toolset for log analysis, threat hunting and building insightful dashboards.

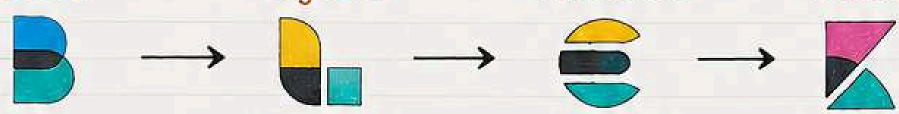
☆ By the end of this room, you will be able to use ELK for investigations and build visualizations to support SOC operations! ☆

Task 2: Elastic Stack Overview

Elastic Stack (ELK) was originally developed to store, search, and visualize large amounts of data. Organizations used it to monitor application performance and perform searches on large datasets. Over time, its features made it popular in security operations as well. Now, many SOC teams use ELK almost as a SIEM solution.

> Elastic Stack is a collection of different open-source components that work together to collect data from any source, store and search it, and visualize it in real time.

Elastic Stack Components



Before we go on to learning log analysis through ELK, let's first discuss its core components.

① Elasticsearch

Elasticsearch is a full-text search and analytics engine for JSON-formatted documents. It stores, analyzes, and correlates data and supports a RESTful API for interacting with it.

② Logstash

Logstash is a data processing engine that takes data from different sources, filters it, or normalizes it, and then sends it to the destination, which could be Kibana or a listening port.

A Logstash configuration file is divided into three parts:

Input	Filter	Output
Where the user defines the source from which the data is being ingested.	Where the user specifies the filter options to normalize the log ingested above.	Where the user wants the filtered data to be sent.

It can be a listening port, Kibana Interface, Elasticsearch database, or file.

- Logstash supports many Input, Output, and Filter plugins.

③ Beats

Beats are host-based agents (data-shippers) that ship/transfer data from the endpoints to Elasticsearch. Each beat is a single-purpose agent that sends specific data to Elasticsearch. All available beats are shown below.

④ Kibana

Kibana is a web-based data visualization tool that works with Elasticsearch to analyze, investigate, and visualize data streams in real time. It allows users to create multiple visualizations and dashboards for better visibility. There is more on Kibana in the following tasks.

★ **Note:** ELK is not a traditional SIEM, but many SOC teams use it like one!

How they work together?

Note: As a SOC analyst, your primary responsibility is to work with ELK to perform log analysis and investigations. You do not need to specialize in how each component behind the ELK works, but basic understanding is essential.

How they work together (step-by-step):

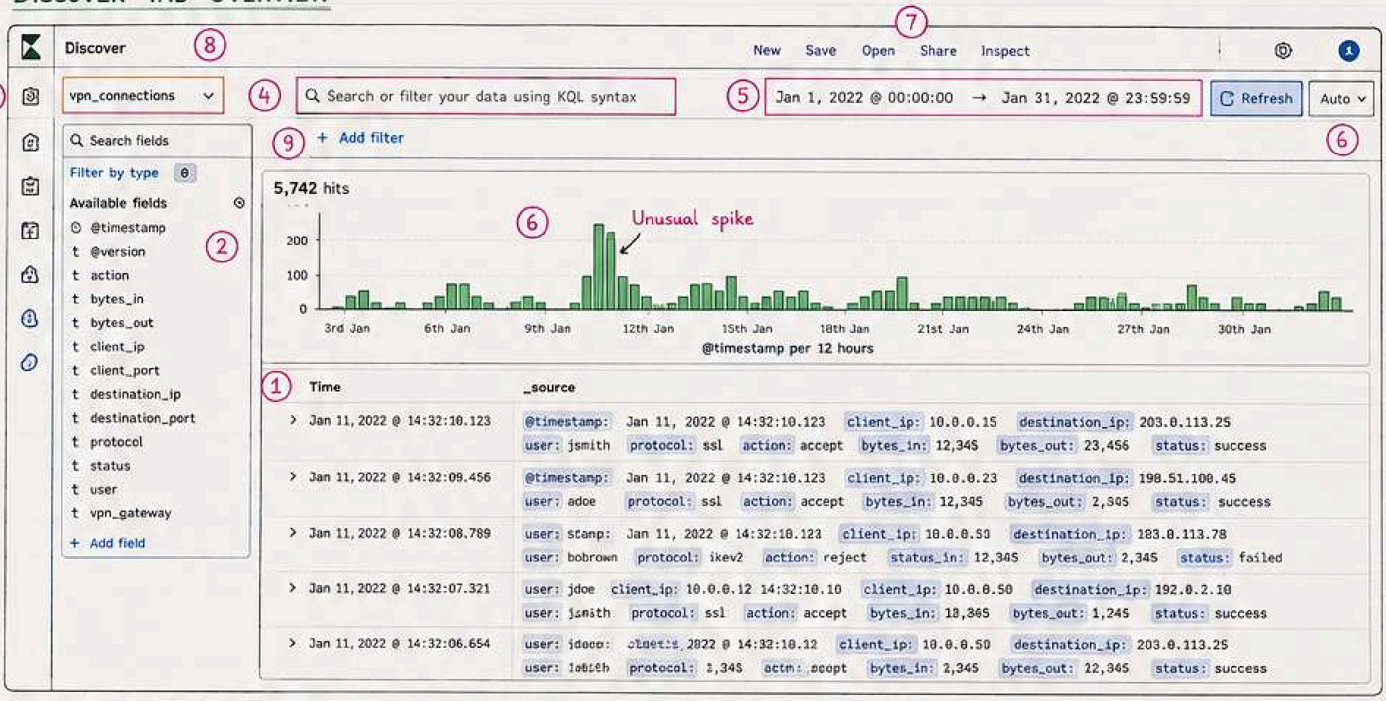
1. Beats collect data from multiple agents. For example, Winlogbeat collects Windows event logs, and Packetbeat collects network traffic flows.
2. Logstash collects data from beats, ports, or files, parses/normalizes it into field value pairs, and stores them into Elasticsearch.
3. Elasticsearch acts as a database used to search and analyze data.
4. Kibana is responsible for displaying and visualizing the data stored in Elasticsearch. The data can easily be shaped into different visualizations, time charts, infogr phics, etc., using Kibana.

Task 4 : Discover Tab

From this task onwards, we will discuss ELK's front-end interface. These are the main features that a SOC analyst operates on. As discussed in the second task of this room, Kibana is the component of ELK that supports these interactions with the front end.

Note: The Discover tab is where SOC analysts spend most of their time. It is the main workspace for exploring, searching and analyzing raw data.

Discover Tab Overview



What Each Element Does

1. **Logs** – Each row shows a single log containing information about the event, along with the fields and values.
2. **Fields Pane** – The left panel shows the list of fields parsed from the logs. Click any field to add it to the filter or remove it from the search.
3. **Index Pattern** – Each type of log is stored in a different index pattern. Select the index pattern from which we need the logs (e.g., VPN logs → vpn_connections).
4. **Search Bar** – Add search queries and apply filters to narrow down the results.
5. **Time Filter** – Narrow down results based on any specific time duration.
6. **Time Interval** – This chart shows the event counts over time.
7. **TOP Bar** – Contains options to save the search, open saved searches, share or save, etc.
8. **Discover Tab** – Main workspace in Kibana for exploring, searching and analyzing raw data.
9. **Add Filter** – Apply filters to specific fields to narrow down results, instead of typing entire queries.

Tip: Use filters and time range together to quickly find the data you need!

Important Elements Explained

1) Index Pattern

By default, Kibana requires an index pattern to access the data stored in Elasticsearch. It tells Kibana which Elasticsearch data we want to explore. Each index pattern corresponds to certain defined properties of the fields. A single index pattern can point to multiple indices.

Each log source has a different structure; therefore, logs are first normalized into fields and values by creating a dedicated index pattern.

In this lab, we will explore the index pattern: vpn_connections (VPN logs)

2) Fields Pane

Shows the list of normalized fields found in the logs. Click any field – it shows top 5 values and % of occurrence.

Example:

client_ip

Top 5 values	%	
10.0.0.15	35%	+
10.0.0.23	20%	+
10.0.0.50	15%	+
10.0.0.12	10%	+
10.0.0.33	8%	−

[+] Add filter (include)

[−] Add filter (exclude)

3) Time Filter

Allows us to apply a log filter based on time.

Quick options:

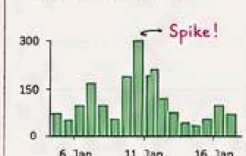
- Last 15 minutes
- Last 1 hour
- Last 24 hours
- Last 7 days
- Last 30 days
- Last 15 years
- Custom (choose dates)

4) Timeline

Provides an overview of the number of events that occurred over time.

Select any bar to show logs in that period.

The count at top left shows total events.



Helps in identifying spikes in logs.

5) Create Table

By default, logs are shown in raw form. Click on any log and select important fields to create a table showing only those fields.

@timestamp	client_ip	user	status
11 Jan 14:32:10	10.0.0.15	jsmith	success
11 Jan 14:32:09	10.0.0.23	adoe	success
11 Jan 14:32:08	10.0.0.15	bobrown	failed
...	...	...	

Reduces noise and makes data more meaningful!

Save the table format to use it every time.

TIME RANGE CHECK: Before answering the data questions, set the Kibana time picker to include January 2022. A wide range such as Last 15 years also works for this lab.

★ Task 5 : KQL Overview

> ★ **Note:**
> KQL (Kibana Query Language) is a search query language used to search the ingested logs / documents in Elasticsearch.

Remember the Search Bar we saw in the Discover Tab in the previous task? We can find anything using this option. Let's see how.

There is a special language that we can use inside this search bar to perform our searches. KQL (Kibana Query Language) is a search query language used to search the ingested logs / documents in Elasticsearch.

With KQL, we can search for the logs in two different ways.

- Free text search
- Field-based search

1. Free text Search

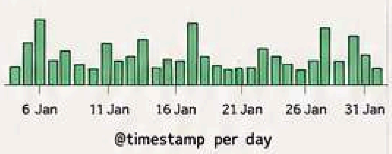
Free text search allows users to search for logs based on text only. That means a simple search of the term United States will return all the documents that contain this term, irrespective of the field.

- Search Query: United States

> What if we only search for the term United?

United States | Last 15 years | Refresh

2,304 hits



Time	_source
› Jan 11, 2022 @ 14:32:10.123	country : United States state : Virginia ... user : jsmith event : vpn_login status : success
› Jan 11, 2022 @ 14:35:21.456	country : United States state : Texas ... user : mdoe event : vpn_disconnect status : success
› Jan 11, 2022 @ 14:36:41.789	country : United States state : New York ... user : jdoe event : vpn_login status : success

It didn't return any results because KQL looks for the whole term/word in the documents.

KQL allows the wildcard * to match parts of the word.

- Search Query: United* (Returns all results containing United and any other term after it)

2. Logical Operators (AND | OR | NOT)

KQL also allows users to utilize logical operators in the search query. Let's look at the examples below.

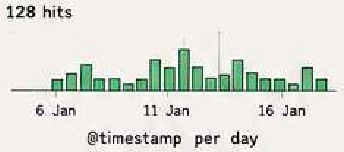
1) AND Operator

Returns the logs containing both "United States" and "Virginia".

- Search Query:

"United States" AND "Virginia"

128 hits



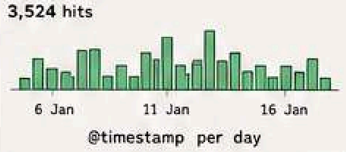
2) OR Operator

Shows logs that contain either "United States" or "England".

- Search Query:

"United States" OR "England"

3,524 hits



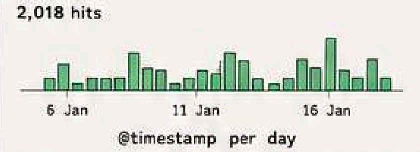
3) NOT Operator

Shows logs from the United States, including all states, but ignoring Florida.

- Search Query:

"United States" AND NOT ("Florida")

2,018 hits



3. Field-Based Search

In the Field-based search, we will provide the field name and the value we are looking for in the logs. This search has a special syntax as Field : Value. It uses a colon as a separator between the field and the value. Let's look at a few examples.

- Search Query: Source_ip : 238.163.231.224 AND UserName : Suleman

  Explanation: We are telling Kibana to display all the logs in which the field Source_ip contains the value 238.163.231.224 and UserName is Suleman.

Source_ip : 238.163.231.224 AND UserName : Suleman | Last 15 years | Refresh

42 hits

Time	_source
Jan 11, 2022 @ 14:42:11.123	Source_ip : 238.163.231.224 UserName : Suleman event : vpn_login status : success ... country : United States state : Virginia
Jan 11, 2022 @ 14:48:21.987	Source_ip : 238.163.231.224 UserName : Suleman event : vpn_logout status : success ... country : United States state : Virginia

> When we click on the search bar, we are presented with all the available fields that we can use in our search query.

Popular fields

- t Source_ip
- t UserName
- t country
- t state
- t event
- t status
- t @timestamp
- t client_ip
- ...

> ☆ Tip:
> - KQL is case-insensitive. (united = United)
> - Use double quotes (" ") for exact phrases.

Task 6: Creating Visualizations

The visualization tab allows us to visualize the data in different forms such as tables, pie charts, bar charts, etc. This visualization task will use the multiple options this tab provides to create some simple presentable visualizations.

> ★ Note: Visualizations help turn raw log data into meaningful insights that are easier to understand, share and act on.

1. Create Visualization

There are a few ways to navigate to the visualization tab. One way is to click on any field in the Discover tab and click on the visualization icon as shown below.

client_ip ⌄ 10.0.0.15 Visualize

We can create multiple visualizations by selecting options like tables, pie charts, bar charts, etc.

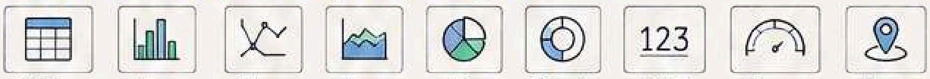


Table | Bar | Line | Area | Pie | Donut | Metric | Gauge | Maps

2. Correlation Option

Often, we require creating correlations between multiple fields. Dragging the required field in the middle will create a correlation tab in the visualization tab. Here, we selected the Source_Country as the second field to show a correlation among the client Source_IP.

Visualize / Correlation

Source fields: client_ip ✕ | Add or drag-and-drop a field

Source_Country ✕

Drag field here to create correlation

Metrics: Count

client_ip vs. Source_Country

	United States	India	United Kingdom	Other	Total
10.0.0.15	45	12	5	3	65
10.0.0.23	30	8	7	2	47
10.0.0.50	10	25	4	1	40
10.0.1.12	18	6	3	1	28
10.0.2.10	8	4	2	1	15
Total	111	55	21	8	195

We can also create a table to show the values of the selected fields as columns, as shown below.

client_ip	Source_Country	Count
10.0.0.15	United States	65
10.0.0.23	United States	30
10.0.0.50	India	25
10.0.1.12	United States	18
10.0.2.10	United States	8
...	...	...

The most important step in creating these visualizations is saving them. To do so, click on the save option on the right side and fill in the descriptive values below.

Save visualization ✕

Title: Client IP vs Source Country (Correlation)

Description: Shows count of events grouped by client IP and Source Country.

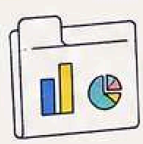
Tags (optional): vpn, correlation, geography

☑ Save and add to library

Cancel | Save

Steps to take after creating Visualizations

1. Create a visualization and click the Save button at the top right corner.
2. Add the title and description to the visualization.
3. Click Save and add to the library when it's done.



Saved visualizations can be used in dashboards later!

3. Failed Connection Attempts Visualization

Failed attempts filter: For the failed connection visualisation, use the vpn_connections data view, set the time picker to include January 2022, then filter for action: failed. Do not exclude failed events; the table should only show failed VPN connection attempts. Use UserName and Source_ip as the table fields.

We'll use the knowledge gained above to create a table to show the user and the IP address involved in failed attempts.

D Discover | action: failed | Jan 1, 2022 @ 00:00:00 → Jan 31, 2022 @ 23:59:59 | Refresh

1,245 hits

UserName	Source_ip
jsmith	203.0.113.25
bobrown	198.51.100.45
jdoe	183.0.113.78
suleman	238.163.231.224
rpatel	203.0.113.25
...	...

> Tip: You can further enhance this table by adding Time, Destination_IP or Protocol as additional columns.

> ★ Visualizations help you quickly identify patterns, summarize information and present insights in a meaningful way.

Task 7 : Creating Dashboards

Dashboards provide good visibility into log collection. A user can create multiple dashboards to fulfill a specific need. In this task, we can combine different saved searches and visualizations to create a custom dashboard for VPN log visibility.

Note: Dashboards give you a single view of important data by combining searches and visualizations that you create and save.

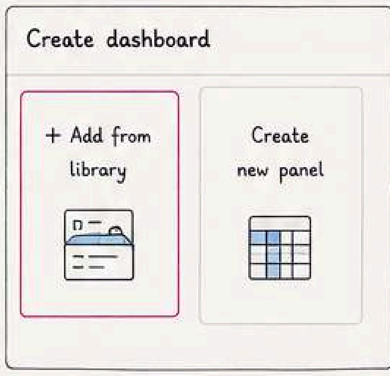
Creating a Custom Dashboard

By now, we have saved a few Searches from the Discover tab, created some Visualizations, and saved them. It's time to explore the dashboard tab and create a custom dashboard.

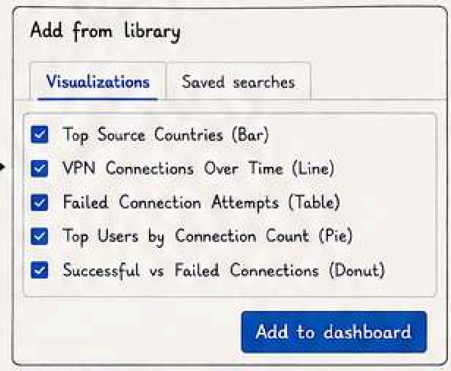
Dashboards help SOC analysts and teams quickly monitor, detect anomalies, and make data-driven decisions.

1. Go to the Dashboard tab and click on the Create dashboard.

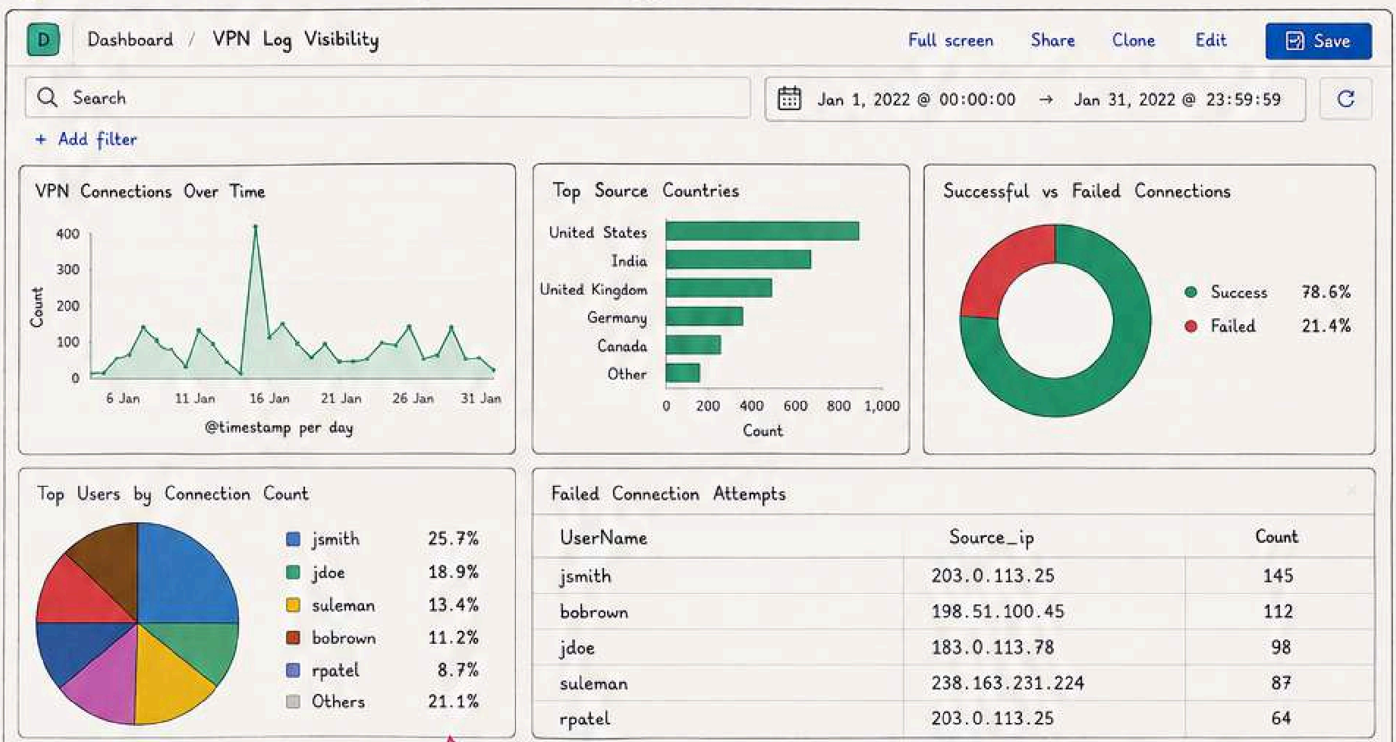
2. Click on Add from Library.



3. Click on the visualizations and saved searches. It will be added to the dashboard.



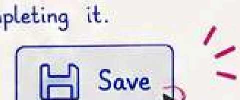
4. Once the items are added, adjust them accordingly, as shown below.



Drag and resize panels to arrange the layout as per your need.

Place important metrics on top and detailed tables at the bottom.

5. Don't forget to save the dashboard after completing it.



Key Takeaways

- ✓ Dashboards combine multiple visualizations and searches.
- ✓ They provide a single view for quick monitoring and analysis.
- ✓ Custom dashboards can be created based on specific needs.
- ✓ Always save dashboards so others can use and benefit from them.